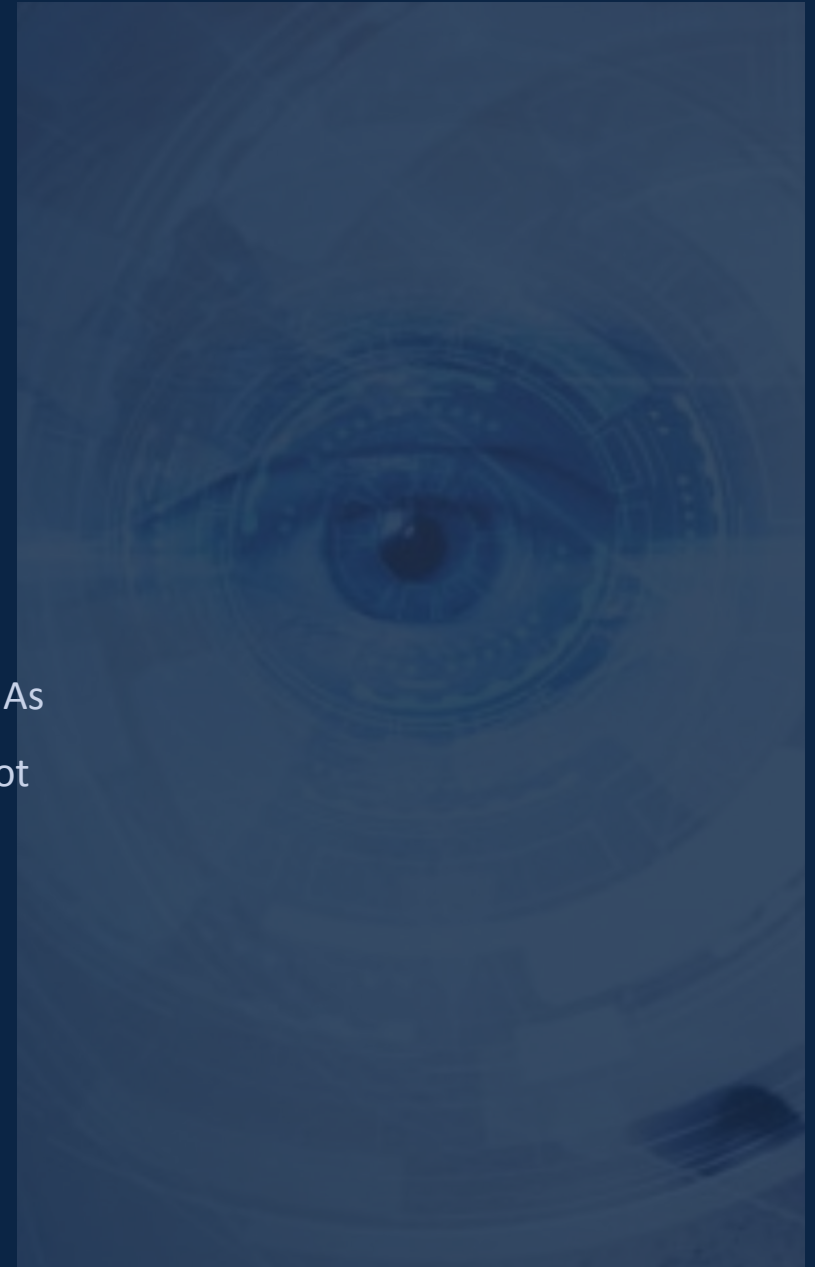


The review queue in a virtual diabetic-retinopathy clinic

Every fundus photograph a virtual clinic captures needs an ophthalmologist's eyes on it. As patient volume grows, that review queue becomes the bottleneck — not the imaging, not the diagnosis capability, the human review time itself.

Growing patient backlog

AI-assisted triage: grade the image, apply clinical rules, draft the report — a clinician confirms every decision. At the operating threshold, the clinician actively reviews the 43% of cases that don't clear automation — not all of them.



15 years of nursing, now applied to the model

1

Augmentation constraints

Vertical flips are clinically impossible for a fundus image — excluded from the pipeline on that basis, not by trial and error.

2

Confidence thresholds

Grad-CAM showed the model struggles most on intermediate grades — the same grades that are genuinely hardest to read clinically. A stricter 0.75 threshold follows from that overlap.

3

Follow-up intervals

The 12/6/6/1/1-month intervals by grade are clinical judgment, not learned or invented — Grade 3's 1-month interval is intentionally stricter than standard screening.

Clinical background didn't just inspire this project — it shaped specific, defensible engineering decisions throughout.

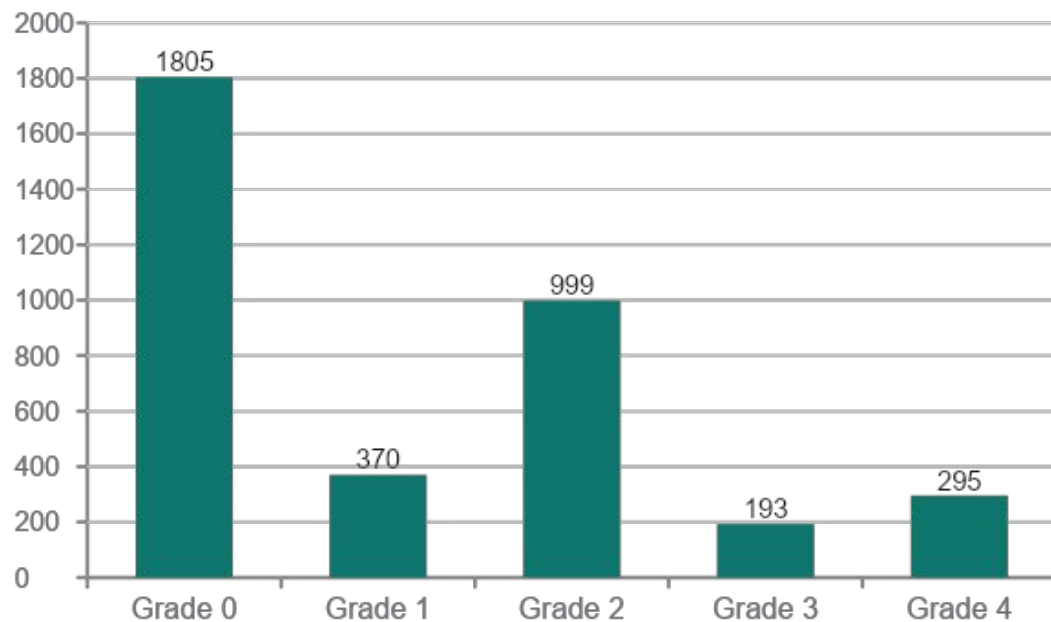
Grading follows the ICDRSS 5-grade scale (Wilkinson et al., Ophthalmology, 2003).

DATASET

APTOS 2019 — 3,662 fundus photographs, 5 severity grades

Originally released as a Kaggle competition by the Asia Pacific Tele-Ophthalmology Society, evaluated by Quadratic Weighted Kappa.

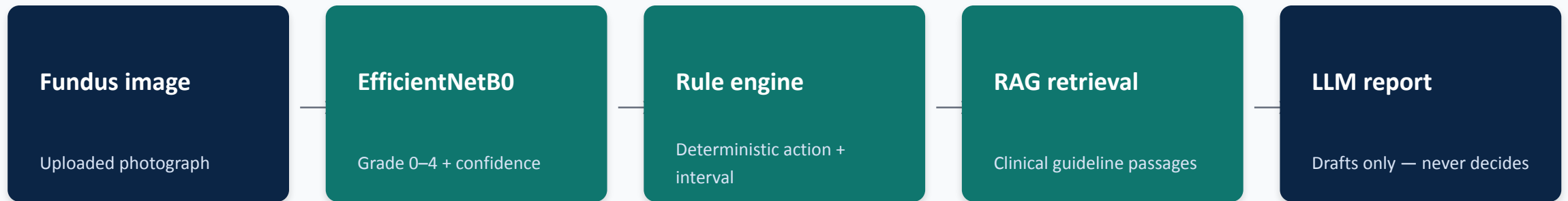
Class distribution



Known limitations, documented upfront

- **9.4 : 1 imbalance**
between Grade 0 and Grade 3 — the class that matters most clinically is the rarest.
- **10+ distinct resolutions**
~24x pixel-area spread across the dataset — no uniform capture standard.
- **Luminance 19.6–106**
a wide exposure range; the lower tail overlaps with visibly unusable captures.
- **Known split leakage**
a limitation in the train/val/test split, documented rather than hidden.

Five stages, each with a clear responsibility

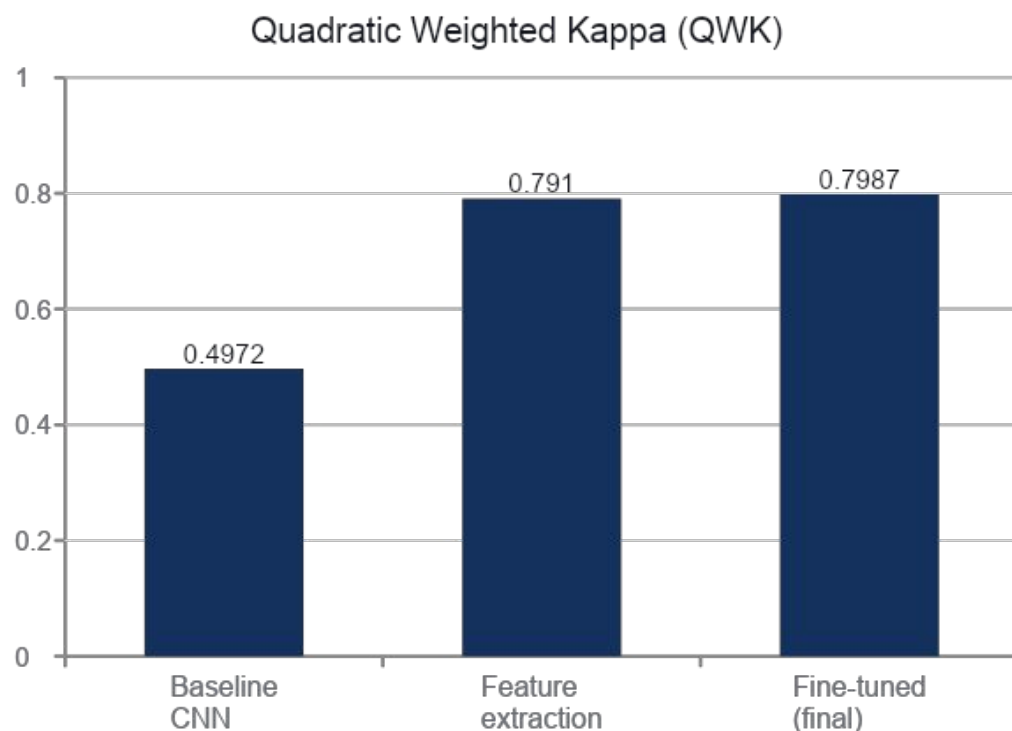


Design principle

Grade, confidence, action, and follow-up interval are always copied directly from the rule engine — never re-read or overridden from LLM output. The LLM drafts the narrative; it never touches the safety-critical fields.

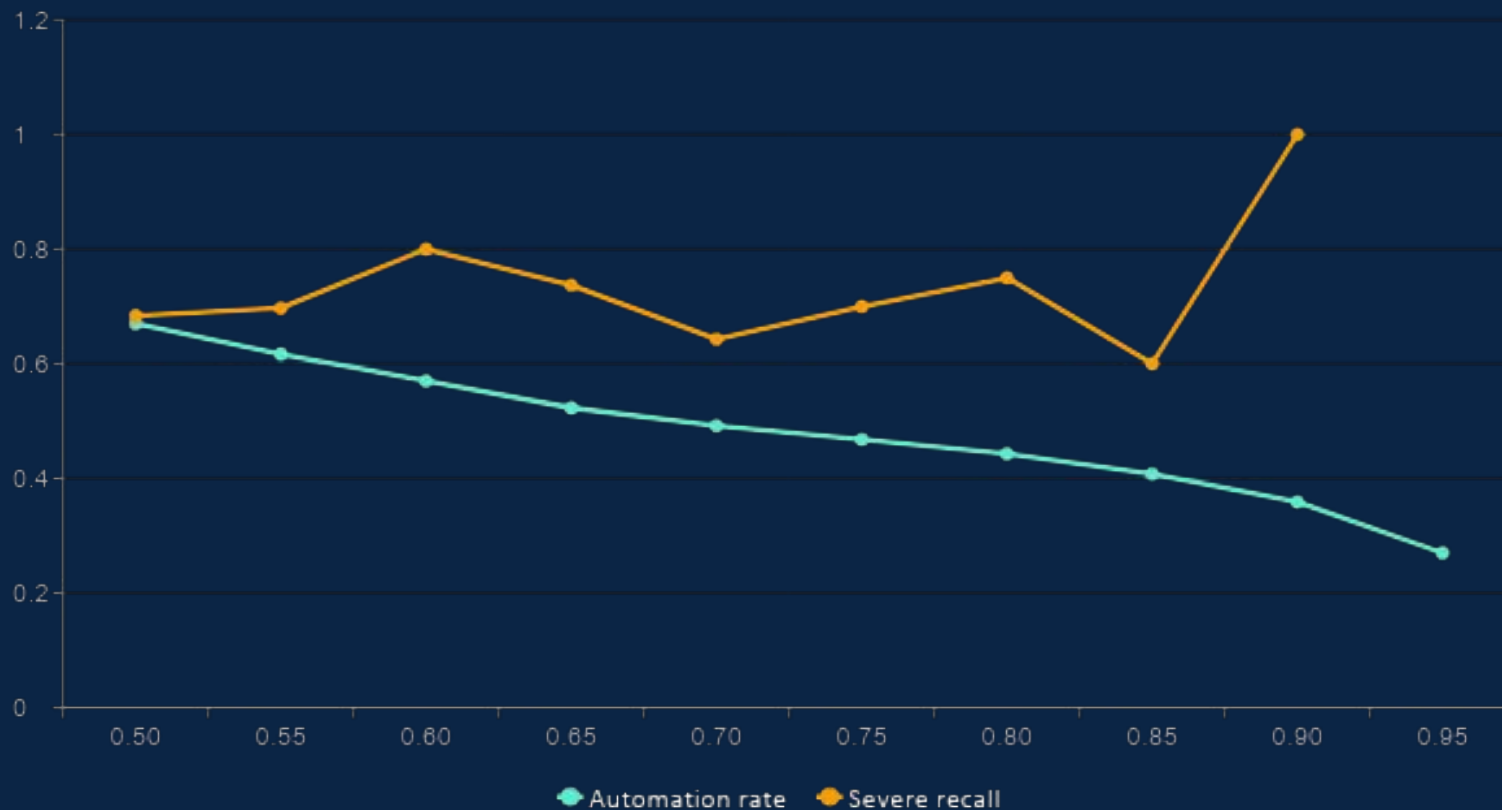
Patient intake (HbA1c, diabetes duration, spherical refraction) feeds the rule engine alongside the predicted grade and confidence.

Fine-tuning won on the metric that matters clinically



Fine-tuning was selected for its Grade 3 recall — catching severe cases matters more than overall accuracy or QWK alone.

Automation vs. severe-case recall: where do you set the threshold?



Operating point: 0.60

Below 0.60, automation climbs fast but severe recall becomes volatile. Above it, automation drops sharply while recall gains flatten out.

0.60 keeps roughly 57% of cases fully automated while holding severe recall at 0.80 — the clearest point on the curve where both sides are still winning.

What the model actually looks at

Grad-CAM finding

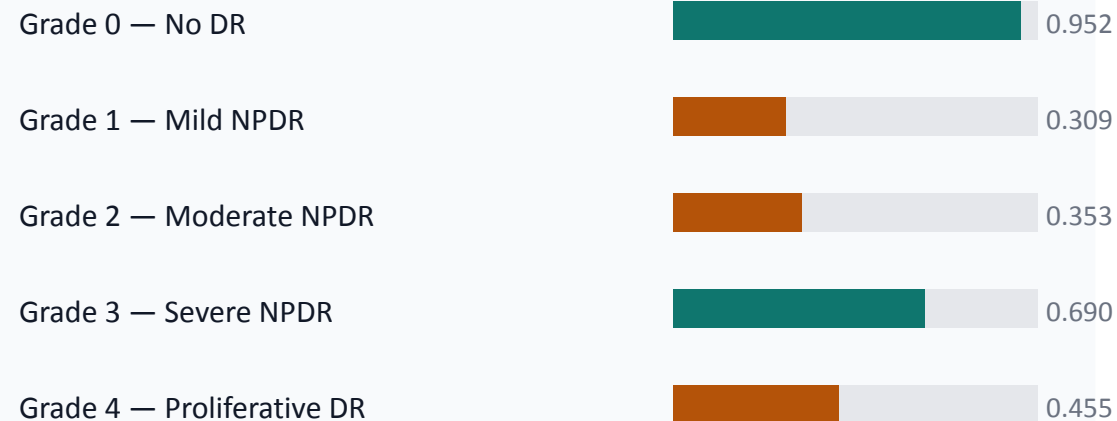
Attention concentrates on the optic disc across all five grades, rather than shifting to grade-specific lesions.

Corroborated clinically

Grade 0 and Grade 4 are easy to tell apart by eye. From a single fundus photograph alone, without other diagnostic tools, the boundary between 1↔2 and 3↔4 is genuinely more subtle — for the model, and for a reader working from that image alone.

This is a finding about task difficulty, documented as such — not a model flaw papered over.

Per-grade recall reflects it



RAG-grounded reports — evaluated, not assumed

100%

Hit rate @3

The correct guideline chunk is retrieved in the top 3, every time in the eval set.

~78%

Hit rate @1

Top-1 alone is lower — the reason retrieval uses top-3, not top-1.

7

Guideline chunks

A deliberately small corpus. A chunk-prefix experiment was tried and reverted — it made retrieval worse.

The LLM drafts. It never decides.

Grade, confidence, action, and follow-up interval are copied directly from the deterministic rule engine into the final report — never re-generated or re-read from LLM output.

Screening → Grad-CAM → structured report

- 1 Upload a fundus photograph, patient intake pre-filled from synthetic data
- 2 Model predicts grade + confidence; Grad-CAM overlay shown alongside the original
- 3 Rule engine applies clinical logic — flags for human review when confidence is low
- 4 RAG retrieves guideline passages; LLM drafts the structured pre-report
- 5 Insights tab: model metrics, this session's automation rate, full Tableau dashboard

```
streamlit run app/streamlit_app.py
```

What this is — and what it isn't, yet

Not a diagnostic device

An educational prototype, not certified or regulator-approved. Every output requires clinician review before being acted on.

Current scope limits

- Only 3 of 5 intake fields (HbA1c, diabetes years, spherical refraction) feed the rule engine — age and visual acuity are captured but not yet modelled
- RAG corpus is intentionally small (7 chunks) — an accepted limitation, not an active optimization target
- Deployment layer (API/Docker) was scoped out — documented, not attempted under time pressure

Future work

- Incorporate age and visual acuity into the rule engine's modifiers
- Expand the guideline corpus and re-evaluate retrieval hit rate
- External validation on a second, independently sourced fundus dataset
- Package the API/Docker deployment layer

Thank you.